In 1D, the WKB approximation for the wavefunction (at order $\hbar^1$) is given by

$$\psi(x) = \frac{1}{\sqrt{p(x)}}\left[Ae^{\frac{i}{\hbar}\int^x p(x')dx'} + Be^{-\frac{i}{\hbar}\int^x p(x')dx'}\right], \tag{1}$$

where $p(x) = \sqrt{2m(E - V(x))}$ is the classical momentum. Note that A and B are constants determined by boundary conditions. They describe left and right moving waves, respectively.

For a bound state, the energy of the particle E is less than the potential at some point, which means that there are classically forbidden regions where $p(x)$ becomes imaginary. The point at which this happens is called a turning point, and the WKB approximation no longer holds right at that intersection. To resolve this, one matches the allowed and forbidden region solutions, which in the case of a bound state, leads to a quantization condition for the wavefunction (that is, we find standing wave solutions when the energy is below the potential barriers). Note that classically, the particle would be trapped between the turning points. In other words, classically, the nodes of the wavefunction must coincide with the turning points. In quantum mechanics, however, the particle can tunnel through the barriers even if the probability density still decays exponentially fast outside of it (the "evanescent wave"), but the wavefunction of the bound state must still describe a standing wave.

The matching procedure and therefore the quantization conditions depend on what potential we have. The case of the infinite square well is, however, the easiest one, so we derive it below. The matching procedure will be different if our sharp walls are finite and different still if the walls are "smooth". Sakurai's discussion is concerned with the latter case, which is solved by assuming the potential can be linearized near the turning points. While a bit messy, the end result is very similar to the previous cases and they all become more and more alike for larger quantum numbers n (of course, they must all asymptote to the classical limit).

HW 3.1: In this question we will derive the quantization condition for the infinite square well and derive its energy levels.

a) Consider the infinite square well potential

$$V(x) = \begin{cases} V_0 & \text{if } 0 < x < a, \\ \infty & \text{otherwise.} \end{cases} \tag{2}$$

What are the turning points of this potential? Using the wavefunction form in Eq. (1) and the boundary conditions for a infinite square well, show that the quantization condition for this potential is just

$$\int_{x_0}^{x_1} dx\, p(x) = n\pi\hbar, \quad n = 1, 2, 3, ... \tag{3}$$

where the integration is taken over a half cycle of the particle, which is defined as the path between one turning point, x_0, to the other, x_1. You can choose either complex exponentials or trig functions to solve this problem.

b) Use the quantization condition to derive the energy levels of the infinite square well with $V_0 = 0$, and compare with the exact solution (namely, $E_n = n^2\pi^2\hbar^2/2ma^2$).

c) Recall we used defined the quantum potential in our notes: $Q(x) = -\frac{\hbar^2}{2m}\frac{1}{\sqrt{p(x)}}\frac{d^2}{dx^2}\sqrt{p(x)}$. Calculate $Q(x)$ inside the infinite square well potential.

d) Imagine now that instead of a flat well ($V(x) = V_0$), we have an arbitrary but continuous potential that satisfies $V(x) < E$ inside the infinite well. Do we expect the WKB approximation to be exact in this case? If not, under what conditions do we expect the WKB approximation to work well?

HW 3.2: Now we look into a couple of other potentials. Given a quantization condition, we will find the turning points and then derive the energy levels of the system.

a) Consider the smooth wall version of the infinite square well, given by the potential

$$V(x) = \begin{cases} V_0 & \text{if } -a < x < a, \\ V_0 \left(\frac{x}{a}\right)^\gamma & \text{if } x \geq a, \\ V_0 \left(\frac{-x}{a}\right)^\gamma & \text{if } x \leq -a, \end{cases} \tag{4}$$

where $V_0 < E$ is a positive constant and γ is a positive integer. This potential has the same shape as the infinite square well, but with smoother walls. Because of it, the nodes of the wavefunction are inside the potential and the quantization condition becomes (same as Eq. 2.254 in Sakurai)

$$\int_{x_0}^{x_1} p(x)dx = \left(n + \frac{1}{2}\right)\pi\hbar, \quad n = 0, 1, 2, ... \tag{5}$$

Find the turning points x_0 and x_1 of this potential. Use this quantization condition to derive the energy levels of this potential, and compare with the infinite square well case.

b) Finally, consider an atom with a single electron, with

$$V(r) = -\frac{\alpha Z}{r}, \tag{6}$$

where α is the fine structure constant and Z is the atomic number. We can imagine $r = 0$ as a hard wall (first turning point) and the second turning point to be some value a such that $E = V(a)$, that is $a = -\frac{\alpha Z}{E}$. The quantization condition for this potential is given by

$$\int_{x_0}^{x_1} p(r)dr = \left(n - \frac{1}{4}\right)\pi\hbar, \quad n = 1, 2, ... \tag{7}$$

Find the turning points x_0 and x_1 of this potential. Use this quantization condition to derive the energy levels of the atom. Do we expect the solution for $n > 1$ to be a good approximation for the Hydrogen atom energy levels? How about the $n = 1$? Compare the latter with the exact solution for ground state of the hydrogen atom.